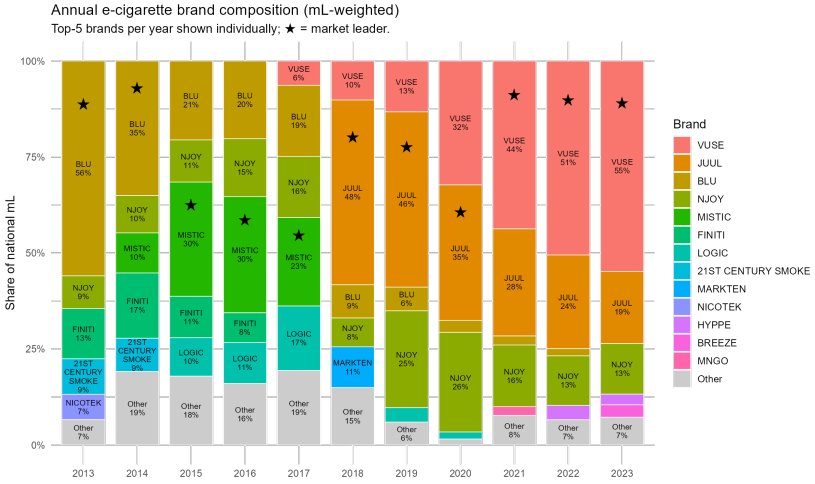
A Game of Leapfrog: Regulation and Market Movements in the U.S. E-Cigarette Market

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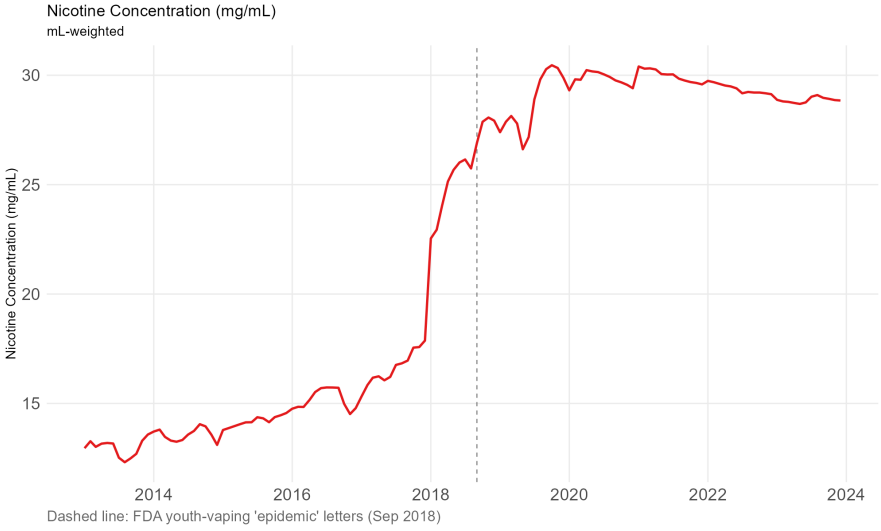
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Motivation



- Small-N, leadership turns over, "Other" (the fringe) is non-negligible

Nicotine concentration has risen



FDA interventions 2018+

2018 FDA enforcement begins — youth-vaping "epidemic" declared; flavor-restriction plan announced (Nov)

2019 EVALI health scare (fall); JUUL pulls fruit/mint flavors (Nov)

2020 Flavored-cartridge enforcement takes effect (Feb); market authorization requirement (Sep)

2022 Synthetic-nicotine loophole closed; FDA denies JUUL's marketing authorization (Jun)

Counterfactuals

Absent this escalation, how would:

- Nicotine concentration (mg/mL) have evolved?
- Market leadership have evolved?

The plan

Today

Static pricing, **exogenous** quality state, dynamic entry & exit.

Future

Adds **endogenous** investment (binary invest/don't-invest), FDA regulation entering through a firm-and-time-varying investment cost.

Data

- Kilts Nielsen Retail Scanner Data
- Weekly store sales, UPC (barcode) level, 2013–2023
- Aggregated to firm $\times$ month

Environment

Players (period t, monthly):

- Up to $\bar{N} = 5$ incumbent firms, $i = 1, \ldots, n_t \leq \bar{N}$
- One explicit competitive fringe F (aggregated)
- At most one potential entrant per period

State: $\omega_{i,t} = (a_{i,t}, \xi_{i,t})$

- $a_{i,t}$: accumulated-quality index — **observed** data. Product-unweighted average nicotine content (mg/mL) across firm i's active products that month
- $\xi_{i,t}$: idiosyncratic residual — **not** observed directly; recovered from the demand estimation
- Both evolve exogenously via first order Markov

$a_{i,t}$ evolution by firm →

Timing

Each period: states realize $\rightarrow$ static pricing subgame $\rightarrow$ profits realize $\rightarrow$ exit/entry decided $\rightarrow$ states evolve.

Demand

Nested logit: incumbents (i) and the fringe (F) share one e-cigarette nest against a degenerate outside-option nest.

$$\delta_{i,t} = \beta_a a_{i,t} + \xi_{i,t} - \alpha p_{i,t}$$

- $a_{i,t}$: observed quality (nicotine concentration, mg / mL)
- $\xi_{i,t}$: exogenous residual

$p_{i,t}$: unweighted mean of UPC real revenue / corrected mL sold.
Reviewed pack volumes/counts update retail demand inputs; NCP input unchanged.

Diagnostics: unweighted cartridge/disposable mL; log eligible UPC count.
Compare national/geographic count IVs, with/without firm FE.
Annual/monthly/constant NCP-intensity market sizes; report size-missing exclusions.

Revenue caveat: representative price times firm mL need not recover revenue.
Current results: comparison report. Retail rebuild: 7 October 2026.

Static pricing subgame

$$\pi_i(\omega_t) = \max_{p_{i,t}} \left(p_{i,t} - c(\omega_{i,t})\right) q_i(p_t, \omega_t) \; - \; FC$$

Bertrand–Nash equilibrium price vector, solved period-by-period given the vector of all firms' states, ω_t.

$q_{i,t} = M_t \cdot s_i(p_t, \omega_t)$: quantity = market size $\times$ predicted share.

Dynamic entry and exit

Incumbent's value function:

$$V_i(\omega_{i,t}, \omega_{-i,t}) = \pi_i^*(\omega_t) + \max\Big\{ \underbrace{\phi_i}_{\text{exit: scrap value}} \;,\; \underbrace{\beta\,\mathbb{E}[V_i(\omega_{i,t+1}, \omega_{-i,t+1}) \mid \omega_t]}_{\text{continue}} \Big\}$$

$\beta \in (0,1)$: discount factor.

Entrant's problem: enter iff

$$\beta\,\mathbb{E}[V_e(\omega_{e,0}, \omega_{-e,t+1}) \mid \omega_t] \;-\; \underbrace{x_e}_{\text{sunk entry cost}} \;\geq\; 0$$

Equilibrium: a **Markov Perfect Equilibrium (MPE)** — a fixed point in the exit/entry cutoff rules and the value function — computed via Pakes–McGuire (1994) iteration.

Pakes–McGuire steps →

The fringe

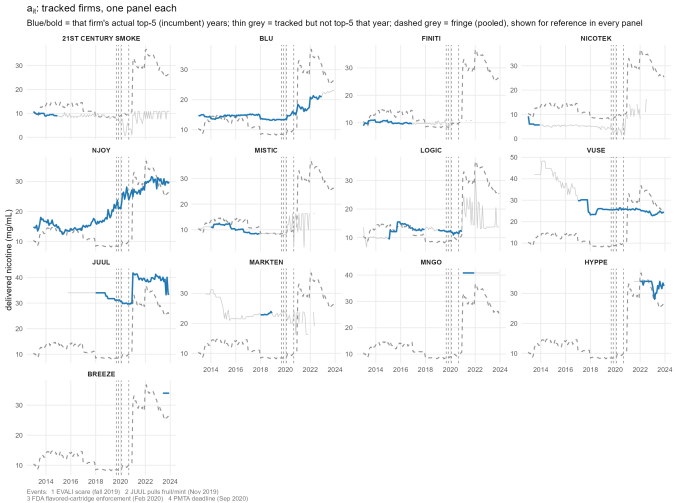
- Model explicitly due to non-negligible market share, and to avoid curse of dimensionality
- Affects incumbents' pricing decisions via entering nested logit demand
- Fringe always exists, therefore no entry/exit decision
- Its state $a_{F,t}$ is constructed directly from the data, the same way as $a_{i,t}$

Looking forward

1. **Endogenous $a_{i,t}$.** Firms choose to invest in quality. Leads to large state space. *MPE* $\rightarrow$ *EBE* Why? $\rightarrow$
2. **Period length.** Monthly (current) vs. quarterly vs. annual — richer regulatory-event timing and more transitions to estimate from, against computational burden.
3. **The fringe.** Market share endogenous.
4. $\bar{N} = 5$ **at capacity.** When the roster is full and a firm wants to enter, does that require a same-period incumbent exit?

Questions

$a_{i,t}$ evolution by firm



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Berry (1994) inversion

With aggregate share data, σ, β_a, and α are recovered from a single linear IV/2SLS regression.

$$\underbrace{\ln(s_{i,t}) - \ln(s_{0,t})}_{\text{log share ratio}} = \underbrace{\sigma \ln(s_{i,t}/s_{g,t})}_{\text{within-nest share}} + \beta_a a_{i,t} - \alpha p_{i,t} + \xi_{i,t}$$

- $s_{i,t}$, $s_{0,t}$, $s_{g,t}$: firm, outside-option, and nest shares — fringe volume enters the denominators here, but the fringe is not its own observation
- Estimated on incumbents only; $\xi_{i,t}$ is the regression residual

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Pakes–McGuire (1994) iteration

Computes a MPE in a dynamic oligopoly model with a large, discrete state space.

1. Guess a value function for each firm.
2. Given that guess, solve each firm's optimal exit and entry thresholds from the Bellman equation.
3. Update the value function by iterating the Bellman equation forward.
4. Repeat until both the policies and the value function converge.

A fixed point in policies *and* values jointly — not solved in closed form. Feasible here because payoffs and transitions are anonymous (depend on the state vector, not on firm identity).

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Why Experience-Based Equilibrium, eventually

Introducing endogenous investment leads to a large state space:

- Up to $\bar{N} = 5$ firms $\times$ 2-dimensional state = up to 10 dimensions
- A firm's best response must be optimal against rivals' *entire* state vector

Fershtman & Pakes (2012), Experience-Based Equilibrium

Relaxes consistency to states likely to happen, and lets firms condition on a coarser summary of rivals rather than the full state vector.

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